

Dirty Paper Coding vs. TDMA for MIMO Broadcast Channels

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Submitted to IEEE Trans. on Information Theory

June 10, 2004

Abstract

We compare the capacity of dirty-paper coding versus that of TDMA for a multiple antenna (MIMO) broadcast channel. We find that the sum-rate capacity (achievable using dirty-paper coding) of the multiple-antenna broadcast channel is at most $\min(M, K)$ times the largest single-user capacity (i.e. the TDMA sum-rate) in the system, where M is the number of transmit antennas and K is the number of receivers. This result is independent of the number of receive antennas and the channel gain matrix, and is valid at all SNR's. We investigate the tightness of this bound in a time-varying channel (assuming perfect channel knowledge at receivers and transmitters) where the channel experiences uncorrelated Rayleigh fading and in some situations we find that the dirty paper gain is upper bounded by the ratio of transmit to receive antennas. We also show that $\min(M, K)$ upper bounds the sum rate gain of successive decoding over TDMA for the uplink channel, where M is the number of receive antennas at the base station and K is the number of transmitters.

Keywords

Broadcast Channel, MIMO Systems, Channel Capacity, Dirty Paper Coding, TDMA

I. INTRODUCTION

In this paper we consider a broadcast channel (downlink or BC) in which there are multiple antennas at the transmitter (base station) and possibly multiple antennas at each receiver (mobile). Dirty-paper coding (DPC) [1,2] is a promising new transmission technique that allows a base station to efficiently transmit data to multiple users at the same time. Dirty paper coding has been shown to achieve the sum-rate capacity (maximum throughput) of the multiple-antenna broadcast channel [1,3–5], and furthermore, it was recently shown that DPC in fact achieves the full capacity region of the MIMO broadcast channel [6]. However, dirty paper coding is a rather new and complicated scheme which has yet to be implemented in practical systems. Current wireless systems such as Qualcomm's High Data Rate (HDR) system [7] use the much simpler technique of time-division multiple-access (TDMA) in which the base transmits to only one user at a time.

Considering the difficulty in implementing dirty-paper coding, a relevant question to ask is the following: How large of a performance boost does dirty-paper coding provide over TDMA in terms of sum-rate? If dirty paper coding is used in a K -user broadcast channel, any rate vector in the K -dimensional dirty paper coding achievable region can be achieved. Similarly, if TDMA is used, any rate vector in the K -dimensional TDMA rate region can be achieved. It is easy to see that the dirty paper coding achievable region is larger than the TDMA rate region. However, defining a meaningful metric that quantifies the difference between two K -dimensional regions for $K \geq 2$ is quite difficult. Viswanathan, Venkatesan, and Huang [8] first investigated the above question by considering different operating points (i.e. rate vectors in the DPC and TDMA rate regions) that are reasonable for cellular systems, and numerically comparing the rates achievable with DPC and TDMA.

In this paper we focus exclusively on the sum rate capacity, or maximum throughput, achievable using DPC and TDMA. This operating point is quite reasonable when users have channels with roughly equivalent quality (i.e. no large SNR imbalances), but may not be as fair for asymmetric channels because we would expect users with higher SNR's to receive a disproportionate fraction of the total data rate. However, the sum capacity is in general an important figure of merit because it quantifies how much total data flow is possible in a broadcast channel.

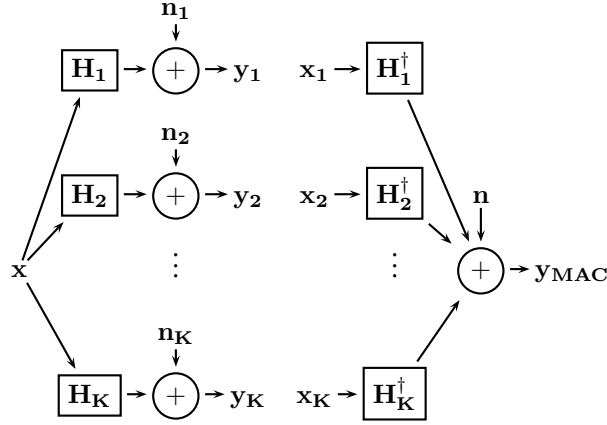


Fig. 1. System models of the MIMO BC (left) and the dual MIMO MAC (right)

Furthermore, comparing the maximum throughput achievable with DPC and TDMA gives a reasonable estimate of how much “larger” the DPC rate region is compared to the TDMA rate region.

By establishing upper and lower bounds to the DPC sum-rate capacity and the maximum TDMA sum-rate, respectively, we are able to analytically upper bound the ratio of sum-rate capacity to the maximum TDMA sum rate. Furthermore, we characterize the DPC gain at asymptotically high and low SNR. We also investigate the DPC gain in a time-varying, Rayleigh-fading channel in which the transmitter and receiver have perfect channel knowledge. Using the same techniques as for the downlink, we are also able to upper bound the sum-rate gain that successive decoding provides over TDMA on the uplink (multiple-access) channel.

The remainder of this paper is organized as follows. In Section II we define our system model, and in Section III we give definitions of the sum rate capacity. In Section IV we develop an analytical bound on the DPC gain and we investigate the asymptotic behavior of the DPC gain at low and high SNR in Section V. We study the behavior of the DPC gain in Rayleigh fading channels in Section VI. In Section VIII we consider the DPC gain in a frequency-selective broadcast channel and in Section IX we briefly compare dirty paper coding to transmitter beamforming, another sub-optimal transmission strategy for the broadcast channel. We end by applying our analytical bounds to the multiple-antenna multiple-access channel in Section X and by stating some conclusions.

II. SYSTEM MODEL

We consider a broadcast channel with K receivers, $M > 1$ transmit antennas, and $N \geq 1$ receive antennas at each receiver. The transmitter sends independent information to each of the K receivers. The system is pictured in the left half of Fig. 1.

Let $\mathbf{x} \in \mathbb{C}^{M \times 1}$ be the transmitted vector signal and let $\mathbf{H}_k \in \mathbb{C}^{N \times M}$ be the channel matrix of receiver k where $\mathbf{H}_k(i, j)$ represents the channel gain from transmit antenna j to antenna i of receiver k . The circularly symmetric complex Gaussian noise at receiver k is represented by $\mathbf{n}_k \in \mathbb{C}^{N \times 1}$ where $\mathbf{n}_k \sim N(0, \mathbf{I})$. Let $\mathbf{y}_k \in \mathbb{C}^{N \times 1}$ be the

received signal at receiver k . The received signal is mathematically represented as

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{x} + \mathbf{n}_k \quad k = 1, \dots, K. \quad (1)$$

The covariance matrix of the input signal is $\mathbf{\Sigma}_x \triangleq \mathbb{E}[\mathbf{x}\mathbf{x}^\dagger]$. The transmitter is subject to an average power constraint P , which implies $\text{Tr}(\mathbf{\Sigma}_x) \leq P$. Throughout the paper, we refer to P as the SNR of the system, though the true received SNR is a function of the channel matrix and the transmission strategy. We let $\mathbf{H} \in \mathbb{C}^{NK \times M}$ refer to the concatenation of all channels, where $\mathbf{H} \triangleq [\mathbf{H}_1^T \cdots \mathbf{H}_K^T]^T$. In the first half of this paper, we assume $\mathbf{H}_1, \dots, \mathbf{H}_K$ are fixed and known perfectly at the transmitter and at all receivers. We explain the time-varying channel model in Section VI.

In terms of notation, we use $\mathbf{H}^\dagger$ to indicate the conjugate transpose of matrix $\mathbf{H}$ and $\|\mathbf{H}\|$ to denote the matrix norm of $\mathbf{H}$, defined by $\|\mathbf{H}\| = \sqrt{\lambda_{\max}(\mathbf{H}^\dagger \mathbf{H})}$. We also use boldface to indicate vector and matrix quantities.

III. SUM-RATE CAPACITY

For the single antenna broadcast channel, the sum rate capacity is equal to the largest single-user capacity in the system [9]. Equivalently, throughput is maximized by transmitting only to the user with the largest channel gain. In fact, the single-antenna Gaussian broadcast channel falls into the class of *degraded* broadcast channels [10], for which the sum rate capacity is always equal to the largest single-user capacity in the system. A broadcast channel is degraded if users can be ordered in terms of the quality of signal they receive from the transmitter. In a degraded broadcast channel, there is always one user who has a stronger channel than every other user. For a single-antenna Gaussian broadcast channel, this is the user with the largest channel norm. Since this user has a stronger channel than any other user, he can decode any codeword intended for any other user, and thus sum rate is maximized by only transmitting to this user in a degraded broadcast channel.

The multiple transmit antenna broadcast channel, however, is a non-degraded broadcast channel. The fact that matrices can only be partially ordered, as opposed to the full ordering possible on scalars, is an intuitive explanation of the non-degraded nature of the multiple-antenna broadcast channel. For example, in a $K = 2, M = 2, N = 1$ channel with $\mathbf{H}_1 = [1 \ 0.5]$ and $\mathbf{H}_2 = [0.5 \ 1]$, it should be apparent that neither channel is absolutely strongest, since $\|\mathbf{H}_1\| = \|\mathbf{H}_2\|$ but clearly $\mathbf{H}_1 \neq \mathbf{H}_2$. It is also not *a priori* clear how sum capacity should be achieved. It turns out, though, that the sum capacity of multiple-antenna broadcast channels is generally strictly larger than the single-user capacity of any of the users in the system, and the sum capacity is achieved by using dirty paper coding to simultaneously transmit to several users [1, 3–5].

Dirty paper coding is a technique that can be used to pre-subtract interference at the transmitter. In a broadcast channel, the transmitted signal $\mathbf{x}$ generally contains information for multiple receivers. While receiver 1 is trying to decode his intended message, the portion of $\mathbf{x}$ not intended for receiver 1 is a form of interference. However, by using dirty paper coding, this interference can be reduced or completely eliminated. Furthermore, it turns out that this transmission strategy is in fact sum-rate capacity-achieving.

From the results in [1, 3–5], the sum rate capacity, denoted by $\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$, can be expressed as:

$$\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) = \max_{\mathbf{\Sigma}_i \geq 0, \sum_{i=1}^K \text{Tr}(\mathbf{\Sigma}_i) \leq P} \sum_{i=1}^K \log \frac{|\mathbf{I} + \mathbf{H}_i(\sum_{j \leq i} \mathbf{\Sigma}_j) \mathbf{H}_i^\dagger|}{|\mathbf{I} + \mathbf{H}_i(\sum_{j < i} \mathbf{\Sigma}_j) \mathbf{H}_i^\dagger|}, \quad (2)$$

where the maximization is over the $M \times M$ positive semi-definite covariance matrices $(\mathbf{\Sigma}_1, \dots, \mathbf{\Sigma}_K)$. The objective function of the maximization is not a concave function of $(\mathbf{\Sigma}_1, \dots, \mathbf{\Sigma}_K)$. This makes the expression in (2) both numerically and analytically difficult to deal with. Fortunately, we are able to with an alternative expression for the sum rate capacity that comes from the dual multiple-access channel, as explained below.

In [3], the dirty paper rate region is shown to be equal to the capacity region of the dual MIMO multiple-access channel (MAC or uplink) with sum power constraint P . The dual uplink is formed by reversing the roles of transmitters and receivers, i.e. converting the transmitter into an M -antenna receiver and converting each receiver into an N -antenna transmitter. The dual MAC is shown in the right side of Fig. 1. The received signal in the dual MAC is given by:

$$\mathbf{y}_{MAC} = \sum_{i=1}^K \mathbf{H}_i^\dagger \mathbf{x}_i + \mathbf{n} \quad (3)$$

where $\mathbf{x}_i \in \mathbb{C}^{N \times 1}$ is the i -th transmitter signal, $\mathbf{H}_i^\dagger$ is the channel of the i -th transmitter, and the noise is the same as in the downlink (i.e. each component is a circularly symmetric unit variance complex Gaussian). Notice that the dual uplink channel matrix of each user is equal to the conjugate transpose of the downlink channel matrix.

Due to the MAC-BC duality, the sum-rate capacity of the MIMO BC is equal to the sum-rate capacity of the dual MAC with sum power constraint P :

$$\begin{aligned} \mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) &= \mathcal{C}_{MAC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \\ &= \max_{\{\mathbf{Q}_i: \mathbf{Q}_i \geq 0, \sum_{i=1}^K \text{Tr}(\mathbf{Q}_i) \leq P\}} \log \left| \mathbf{I} + \sum_{i=1}^K \mathbf{H}_i^\dagger \mathbf{Q}_i \mathbf{H}_i \right| \end{aligned} \quad (4)$$

where each of the matrices $\mathbf{Q}_i$ is an $N \times N$ positive semi-definite covariance matrix. The expression in (4) is the sum-rate capacity of the dual uplink subject to sum power constraint P . Note that (4) is a maximization of a concave function (of the covariance matrices), for which efficient numerical algorithms exist. In this paper, we use the specialized algorithm developed in [11] to calculate $\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$.

The time-division rate region $\mathcal{R}_{TDMA}$ is defined as the set of average rates that can be achieved by time-sharing between single-user transmissions using constant power P :

$$\mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \triangleq \left\{ (R_1, \dots, R_K) : \sum_{i=1}^K \frac{R_i}{C(\mathbf{H}_i, P)} \leq 1 \right\} \quad (5)$$

where $C(\mathbf{H}_i, P)$ denotes the single-user capacity of the i -th user subject to power constraint P . The single-user capacity of a MIMO channel is given by the following expression¹:

$$C(\mathbf{H}_i, P) = \max_{\{\mathbf{Q}_i: \mathbf{Q}_i \geq 0, \text{Tr}(\mathbf{Q}_i) \leq P\}} \log \left| \mathbf{I} + \mathbf{H}_i^\dagger \mathbf{Q}_i \mathbf{H}_i \right|. \quad (6)$$

¹The expression in (6) is the capacity of the point-to-point channel with channel matrix $\mathbf{H}_i^\dagger$. However, the capacity of the dual (or reciprocal) point-to-point channel with channel $\mathbf{H}_i$ is the same [12].

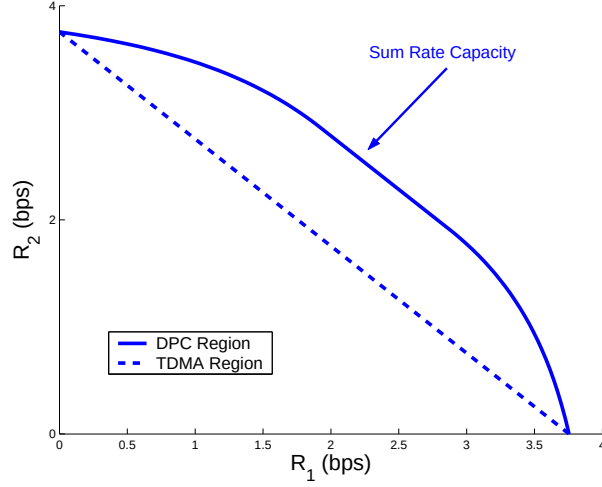


Fig. 2. DPC and TDMA rate regions for a 2 user system with 2 transmit antennas

The maximum is achieved by choosing the covariance matrix $\mathbf{Q}_i$ to be along the eigenvectors of the channel matrix $\mathbf{H}_i \mathbf{H}_i^\dagger$ and by choosing the eigenvalues according to a water-filling procedure [12].

It is easy to see that the maximum sum-rate in $\mathcal{R}_{TDMA}$ is the largest single-user capacity of the K users:

$$\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \triangleq \max_{\mathbf{R} \in \mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)} \sum_{i=1}^K R_i = \max_{i=1, \dots, K} C(\mathbf{H}_i, P) \quad (7)$$

and is achieved by transmitting only to the user with the largest capacity. We will refer to this quantity as the TDMA sum-rate.

In this paper we are interested in quantifying the advantage that DPC gives over TDMA in terms of total throughput. Thus, the performance metric analyzed in this paper is the DPC gain $G(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$, which we define to be the ratio of sum-rate capacity to TDMA sum-rate:

$$G(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \triangleq \frac{\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}. \quad (8)$$

Since $\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \geq \mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ by definition, the DPC gain is always greater than or equal to one. Notice that the DPC gain is a function of the channels $\mathbf{H}_1, \dots, \mathbf{H}_K$ and the SNR P .

In Fig. 2, the DPC and TDMA rate regions are shown for a two user broadcast channel with two transmit antennas and single receive antennas. In this symmetric channel the TDMA sum rate is equal to the single-user capacity of either user (3.75 bps), while the sum rate capacity is equal to 4.79 bps. Thus, the DPC gain is equal to 1.28. In the following section, we develop an analytical upper bound to the DPC gain.

IV. BOUNDS ON SUM-RATE CAPACITY AND DPC GAIN

In this section we develop a precise analytical upper bound to the DPC gain defined in (8). In order to do so, we upper bound the sum-rate capacity of the MIMO BC and lower bound the TDMA sum-rate.

Theorem 1: The sum-rate capacity of the multiple-antenna downlink is upper-bounded by:

$$\boxed{\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \leq M \log \left(1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2 \right)} \quad (9)$$

where $\|\mathbf{H}\|_{max} = \max_{i=1, \dots, K} \|\mathbf{H}_i\|$.

Proof: We prove this result using the fact that the BC sum rate capacity is equal to the dual MAC sum-rate capacity with power constraint P . The received signal in the dual MAC is $\mathbf{y}_{MAC} = \sum_{i=1}^K \mathbf{H}_i^\dagger \mathbf{x}_i + \mathbf{n}$. The received covariance is given by $\Sigma_y = E[\mathbf{y}\mathbf{y}^\dagger] = E[\mathbf{n}\mathbf{n}^\dagger] + \sum_{i=1}^K \mathbf{H}_i^\dagger E[\mathbf{x}_i \mathbf{x}_i^\dagger] \mathbf{H}_i = \mathbf{I} + \sum_{i=1}^K \mathbf{H}_i^\dagger \mathbf{Q}_i \mathbf{H}_i$. Notice that the argument of the maximization in the expression of the sum rate capacity of the dual MAC in (4) is $\log |\Sigma_y|$.

The received signal power is given by $E[\mathbf{y}^\dagger \mathbf{y}] = \sum_{i=1}^K E[\mathbf{x}_i^\dagger \mathbf{H}_i \mathbf{H}_i^\dagger \mathbf{x}_i] + E[\mathbf{n}^\dagger \mathbf{n}]$. Since $\mathbf{x}_i^\dagger \mathbf{H}_i \mathbf{H}_i^\dagger \mathbf{x}_i \leq \|\mathbf{H}_i^\dagger\|^2 \|\mathbf{x}_i\|^2 = \|\mathbf{H}_i\|^2 \|\mathbf{x}_i\|^2$ by the definition of matrix norm, we have

$$E[\mathbf{y}^\dagger \mathbf{y}] \leq \sum_{i=1}^K \|\mathbf{H}_i\|^2 E[\mathbf{x}_i^\dagger \mathbf{x}_i] + E[\mathbf{n}^\dagger \mathbf{n}] \quad (10)$$

$$\leq \|\mathbf{H}\|_{max}^2 \sum_{i=1}^K E[\mathbf{x}_i^\dagger \mathbf{x}_i] + M \quad (11)$$

$$\leq \|\mathbf{H}\|_{max}^2 P + M \quad (12)$$

where (11) follows from the definition of $\|\mathbf{H}\|_{max}$ and the fact that $E[\mathbf{n}^\dagger \mathbf{n}] = M$ and (12) follows from the sum power constraint on the transmitters in the dual MAC (i.e. $\sum_{i=1}^K E[\mathbf{x}_i^\dagger \mathbf{x}_i] \leq P$). Since $E[\mathbf{y}^\dagger \mathbf{y}] = \text{Tr}(E[\mathbf{y}\mathbf{y}^\dagger]) = \text{Tr}(\Sigma_y)$, this implies that $\text{Tr}(\Sigma_y) \leq P \|\mathbf{H}\|_{max}^2 + M$. By [10, Theorem 16.8.4], for any positive definite $M \times M$ matrix $\mathbf{K}$, $|\mathbf{K}| \leq (\frac{\text{Tr}(\mathbf{K})}{M})^M$. Therefore $|\Sigma_y| \leq (1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2)^M$, from which we get $\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) = \max_{\Sigma_y} \log |\Sigma_y| \leq M \log(1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2)$. ■

The upper bound is equal to the sum-rate capacity of a system with M spatially orthogonal eigenmodes (distributed in any manner between the K users), each with norm equal to $\|\mathbf{H}\|_{max}$. Interestingly, users need not be spatially orthogonal for the bound to be achieved with equality. If $N = 1$ and there are more receivers than transmit antennas ($K > M$), then if the users' channels are Welch-bound equality sequences [13] (i.e. $\|\mathbf{H}_i\| = 1$ for all i and $\mathbf{H}^\dagger \mathbf{H} = \frac{K}{M} \mathbf{I}$), then the bound is also met with equality by allocating equal power (choosing $\mathbf{Q}_i = \frac{P}{K}$ in (4)) to each user in the dual MAC. Since $K > M$, it is not possible for the K channels (which are the $1 \times M$ rows of the matrix $\mathbf{H}$) to be mutually orthogonal. However, the Welch-bound condition requires the M columns of $\mathbf{H}$ to be orthogonal. The i -th column of $\mathbf{H}$ refers to the channel gains from the i -th base station antenna to each of the K mobiles.

We now proceed by lower bounding the TDMA sum-rate.

Theorem 2: The TDMA sum-rate is lower bounded by the rate achieved by transmitting all power in the direction of the largest eigenmode:

$$\boxed{\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \geq \log \left(1 + P \|\mathbf{H}\|_{max}^2 \right)} \quad (13)$$

Proof: For each user, $C(\mathbf{H}_i, P) \geq \log(1 + P\|\mathbf{H}_i\|^2)$ because single-user capacity is achieved by water-filling over *all* eigenmodes instead of allocating all power to the best eigenmode. Since the TDMA sum-rate is the maximum of the single-user capacities, the result follows. ■

This bound is tight when $N = 1$, but is generally not tight for $N > 1$ because each user has $\min(M, N)$ eigenmodes to water-fill over.

By combining Theorems 1 and 2, we can upper bound the DPC gain:

$$\frac{\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)} \leq \frac{M \log(1 + \frac{P}{M}\|\mathbf{H}\|_{max}^2)}{\log(1 + P\|\mathbf{H}\|_{max}^2)} \quad (14)$$

$$\leq M \quad (15)$$

where we used Theorems 1 and 2 to get (14). Furthermore, since each user's rate in a broadcast channel can be no larger than his respective single-user capacity,

$$\begin{aligned} \mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) &\leq \sum_{i=1}^K C(\mathbf{H}_i, P) \\ &\leq K \cdot \mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P). \end{aligned} \quad (16)$$

Combining the upper bounds in (15) and (16) gives the following result:

Theorem 3: The DPC gain is upper bounded by M , the number of transmit antennas, and K , the number of users.

$$\boxed{G(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \leq \min(M, K).} \quad (17)$$

This bound is valid for any set of channels $\mathbf{H}_1, \dots, \mathbf{H}_K$, any number of receive antennas N , any number of users K , and any SNR P . When we consider DPC and TDMA from a signaling dimensions perspective, the upper bound is in fact quite intuitive. In the lower bound on TDMA in Theorem 2, only one spatial dimension (corresponding to the largest eigenmode amongst all users) is used. Dirty paper coding, on the other hand, can utilize up to M dimensions² (Theorem 1). Since the TDMA lower bound uses the strongest eigenmode, the quality of each of these M spatial dimensions can be no better than the quality of the dimension used in the TDMA lower bound. Thus the rate on each of the M dimensions can be no larger than the TDMA lower bound, which implies that DPC gives a sum rate no larger than M times the TDMA capacity.

Note: A bound similar to Theorem 1 for the single receive antenna ($N = 1$) downlink when users have the same channel norm and are mutually orthogonal was independently derived in an earlier paper by Viswanathan and Kumaran [14, Proposition 2].

²The term “dimensions” is used rather loosely when applied to DPC. In a single-user non-time varying MIMO channel, spatial dimensions correspond to purely orthogonal signaling directions (generally corresponding to the right eigenvectors of the channel gain matrix). When using DPC, signaling dimensions are generally not orthogonal because orthogonal signaling generally leads to much lower data rates. Thus, the number of spatial dimensions is more accurately interpreted as the rank of the transmit covariance matrix when using DPC.

V. ASYMPTOTIC DPC GAIN

If we consider a system with more transmit antennas than receive antennas ($M \geq N$) in the regimes of high and low SNR, we can establish that the DPC gain converges to $\min\left(\frac{M}{N}, K\right)$ at asymptotically high SNR. Additionally, we can show that the DPC gain converges to unity at low SNR, i.e. that DPC gives no throughput gain at low SNR. We present these results in the following theorems (proofs are given in Appendix I and II):

Theorem 4: If the concatenated channel matrix $\mathbf{H}$ has at least $\min(M, NK)$ linearly independent rows and at least one of the channel matrices $\mathbf{H}_i$ is full row rank (i.e. has N linearly independent rows), as $P \rightarrow \infty$ we have

$$\lim_{P \rightarrow \infty} G(\mathbf{H}_1, \dots, \mathbf{H}_K, P) = \min\left(\frac{M}{N}, K\right). \quad (18)$$

Theorem 5: For any channel matrix $\mathbf{H}$, dirty paper coding and TDMA are equivalent at asymptotically low SNR:

$$\lim_{P \rightarrow 0} G(\mathbf{H}_1, \dots, \mathbf{H}_K, P) = 1. \quad (19)$$

The DPC gain of $\min\left(\frac{M}{N}, K\right)$ at high SNR can be intuitively explained from a dimension counting argument as follows. At high SNR, it is easy to show that the capacity of a MIMO channel grows as $L \log(P)$ (ignoring additive constants), where L is the number of spatial dimensions available in the channel, which is equal to the rank of the channel matrix. Therefore, only the number of spatial dimensions is important at high SNR and the quality of these spatial dimensions (i.e. the channel gain) is unimportant. This is expected since as P becomes large, $\log(1 + \alpha P) \approx \log(P) + \log(\alpha) \approx \log(P)$. When using TDMA, there are $\min(M, N)$ dimensions available. This is equal to the number of dimensions available in an M transmit, N receive antenna MIMO channel. When using DPC, each linearly independent row (i.e. each received signal that is not equal to a linear combination of some other received signals) gives a spatial signaling dimension. Assuming $\mathbf{H}$ has at least $\min(M, NK)$ linearly independent rows, DPC can utilize $\min(M, NK)$ spatial dimensions. When $M \geq N$, the ratio of spatial dimensions using DPC vs. TDMA is given by: $\frac{\min(M, NK)}{N} = \min\left(\frac{M}{N}, K\right)$, and thus DPC gives a sum rate equal to $\min\left(\frac{M}{N}, K\right)$ times the sum rate in TDMA.

Since the sum rate capacity grows as $\min(M, NK) \log(P)$ for large P , it is also easy to show that the sum rate capacity upper bound given in Theorem 1 is asymptotically tight when $M \leq NK$, or that

$$\lim_{P \rightarrow \infty} \frac{\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{M \log\left(1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2\right)} = 1. \quad (20)$$

Interestingly, the TDMA sum capacity lower bound in Theorem 2 is a factor of N times smaller than the actual TDMA sum capacity in the asymptotic limit:

$$\lim_{P \rightarrow \infty} \frac{\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{\log(1 + P \|\mathbf{H}\|_{max}^2)} = N. \quad (21)$$

However, this factor of N does not preclude the general upper bound of $\min(M, K)$ on the DPC gain.

We can also compare the high SNR behavior of the sum rate capacity with that of the cooperative-receiver channel. Since receiver cooperation can only increase the capacity of the broadcast channel, the sum capacity

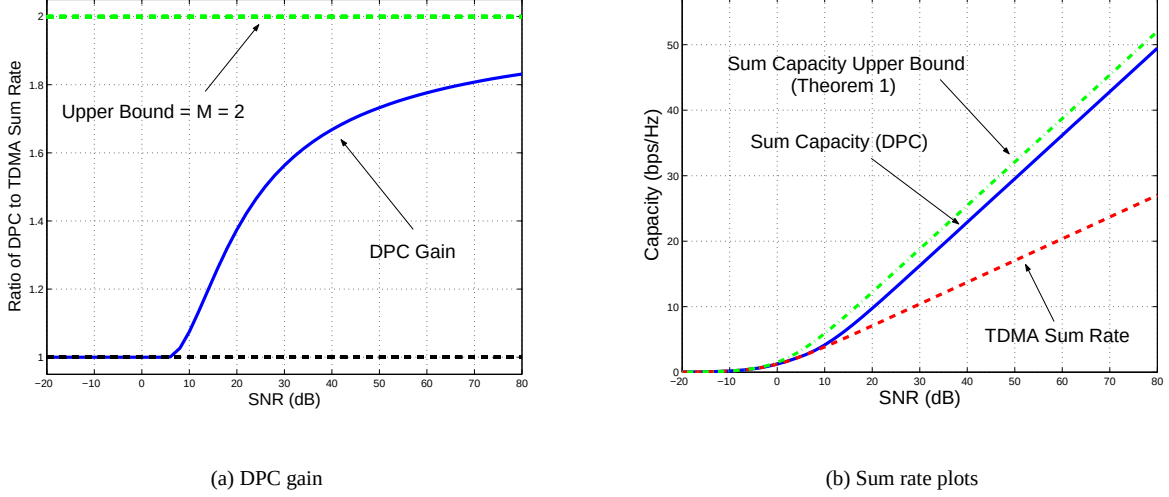


Fig. 3. Plots of DPC gain and sum rate for a 2 user, 2 transmit antenna, 1 receive antenna channel

$\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ is upper bounded by the capacity of the system in which the K receivers fully cooperate, i.e. the capacity of the M transmit, NK receive antenna MIMO channel, which is given by $C(\mathbf{H}, P)$ as defined in (6). The cooperative upper-bound MIMO channel also has $\min(M, NK)$ spatial dimensions. Thus, at high SNR the ratio of sum capacity of the broadcast channel to the capacity of the cooperative channel converges to unity. In fact, an even tighter result shows that the difference between the sum capacity with DPC and the cooperative upper bound converges to zero as SNR goes to infinity [1, Theorem 3].

The intuition behind the low SNR result in Theorem 5 is exactly the opposite of the high SNR scenario. At low SNR only the quality of the best signaling dimension is important and the number of available signaling dimensions is unimportant. To see this, note that for P small, $\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \leq M \log \left(1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2 \right) \approx M \frac{P}{M} \|\mathbf{H}\|_{max}^2 = P \|\mathbf{H}\|_{max}^2 \approx \mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$. DPC allows for simultaneous transmission over all of the different spatial dimensions. Since only the best signaling dimension is of importance at low SNR, this option is of no use and DPC and TDMA are equivalent at low enough SNR.

In Fig. 3(a), the DPC gain $G(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ is plotted as a function of SNR for a 2-user channel with 2 transmit antennas and single receive antennas. The upper bound ($= M = 2$) on the DPC gain is also included. Notice the monotonicity of the DPC gain and convergence of the gain at both low and high SNR. When $N = 1$, we conjecture that $G(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ is in fact a monotonically non-decreasing function of P . Showing that $\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ and $\mathcal{C}_{TDMA}(\mathbf{H}, P)$ are increasing functions of P is trivial, but it appears difficult to show the monotonicity of the ratio of these quantities for even the 2-user, $N = 1$ scenario for which an exact expression for $\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ is known [1, Theorem 1]. In Fig. 3(b), the DPC and TDMA sum rate ($\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ and $\mathcal{C}_{TDMA}(\mathbf{H}, P)$, respectively) are plotted for the same system, along with the sum rate capacity upper bound $M \log \left(1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2 \right)$ from Theorem 1. Note that the sum rate capacity upper bound becomes tight (in the ratio sense only) as $P \rightarrow \infty$. Furthermore, notice that the slope of the sum rate capacity curve is approximately twice

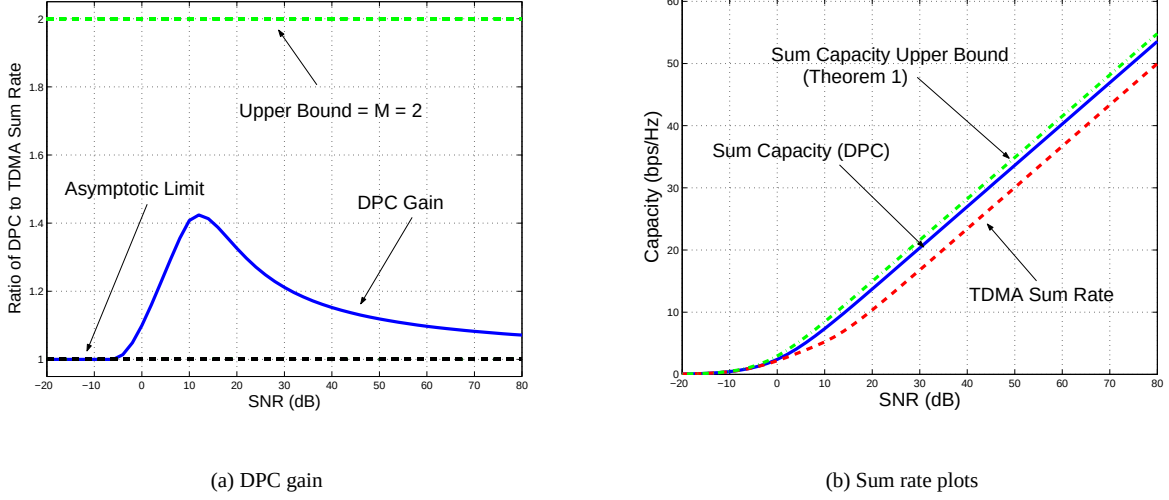


Fig. 4. Plots of DPC gain and sum rate for a 2 user, 2 transmit antenna, 2 receive antenna channel

the slope of the TDMA sum rate curve, leading to the convergence of the DPC gain to two.

In Fig. 4(a), the DPC gain is plotted as a function of SNR for a 2-user channel with 2 transmit antennas and 2 receive antennas per user. Notice that the DPC gain converges to $\frac{M}{N} = 1$ at both extremes, but takes its maximum at a finite SNR. When $M = N > 1$, $G(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ is generally not non-decreasing and actually achieves its maximum at a finite SNR. When $M = N$ as in the figure, both TDMA ($\min(M, N) = M$) and DPC ($\min(M, NK) = M$) can use M spatial dimensions. When using TDMA, the transmitter must choose to use one of the K user's M spatial dimensions. When using DPC, the transmitter can choose M spatial dimensions from all NK available dimensions instead of being forced to choose one of the K sets (corresponding to each user) of N dimensions. This is not important at high SNR, where only the number of dimensions is relevant, or at low SNR, where only the strongest spatial dimensions is relevant). However, this improvement in the quality of dimensions leads to a strict DPC gain at finite SNR. In Fig. 4(b) the DPC and TDMA sum rate are plotted for the same system, along with the sum rate capacity upper bound from Theorem 1. In this channel, it is easy to see that all three curves have the same growth rate, i.e. are asymptotically equivalent in the ratio sense.

VI. TIGHTNESS OF BOUND IN RAYLEIGH FADING

In this section we consider the downlink sum-rate capacity in uncorrelated Rayleigh fading, i.e. where each entry of $\mathbf{H}_k$ is independently and identically distributed as a complex circularly symmetric Gaussian with unit variance. Here we consider a time-varying system, but we assume the transmitter and receiver have perfect and instantaneous channel state information (CSI), and thus can adapt to the channel in each fading state. We also assume that the transmitter (the base station) is subject to a short-term power constraint, so that the base station must satisfy power constraint P in every fading state. This implies that there can be no adaptive power allocation over time³. Assuming

³If the transmitter is subject to an average power constraint instead of a peak power constraint, the fading channel is theoretically equivalent to the frequency-selective broadcast channel model discussed in Section VIII. In the frequency-selective scenario, we show that the DPC gain

that the fading process is ergodic, the sum-rate is equal to the expected value of the sum-rate in each fading state. Therefore, a reasonable performance metric is the ratio of the average sum rate using DPC to the average sum rate using TDMA, i.e. $\frac{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}$. Note that this is not the same as the quantity $\mathbb{E}_{\mathbf{H}}[G(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]$, which is not as meaningful when considering average rates achievable in a fading channel.

In the previous sections we were able to establish bounds and asymptotic limits for the DPC gain for a fixed channel $\mathbf{H}$. In this section, we attempt to gain some intuition about the “average” DPC gain, where we average rates over Rayleigh fading channels and then calculate the ratio. By Theorem 3, we have $\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \leq \min(M, K) \cdot \mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ for each instantiation of $\mathbf{H}$. By taking the expectation of both sides, we get

$$\frac{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]} \leq \min(M, K). \quad (22)$$

In this section we show that this bound can be tightened to $\min(\frac{M}{N}, K)$ in the limit of high SNR, in the limit of a large number of transmit antennas, and in the limit of a large number of users. Note that the same limiting behavior occurs for the DPC gain of each channel instantiation in the limit of high SNR (Theorem 4). We also provide numerical results that show the DPC gain for non-asymptotic systems. To compute the sum rate capacity for each channel instantiation, we use the algorithm provided in [11]. Furthermore, we use the standard Monte Carlo method to approximate the expected value of sum rate over the distribution of $\mathbf{H}$.

A. High SNR

We first consider the scenario where M , N , and K are fixed, but the SNR P is taken to infinity. Furthermore, we assume $K \geq M \geq N$, which is quite reasonable for practical systems. In this scenario, the DPC gain is shown to asymptotically equal $\frac{M}{N}$, or that:

$$\lim_{P \rightarrow \infty} \frac{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]} = \frac{M}{N}. \quad (23)$$

We show convergence by establishing upper and lower bounds on TDMA and DPC sum-rate⁴.

Similar to Theorem 1, we can upper bound the single-user capacity $C(\mathbf{H}_i, P)$ by $N \log(1 + \frac{P}{N} \|\mathbf{H}_i\|^2)$. Then, using Jensen’s inequality, the TDMA sum-rate can be bounded as:

$$\begin{aligned} \mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)] &\leq N \mathbb{E}_{\mathbf{H}} \left[\log \left(1 + \frac{P}{N} \|\mathbf{H}\|_{max}^2 \right) \right] \\ &\leq N \log \left(1 + \frac{P}{N} \mathbb{E}_{\mathbf{H}}[\|\mathbf{H}\|_{max}^2] \right). \end{aligned}$$

is also upper bounded by $\min(M, K)$.

⁴Note that the high SNR result in Theorem 4 cannot be extended in a straightforward manner here because the theorem only gives convergence for each instantiation of $\mathbf{H}$ and we require some uniformity of convergence for the result to hold in an expected value sense across instantiations.

Since $\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \geq \mathcal{C}(\mathbf{H}_1, P)$, we can lower bound the TDMA capacity as:

$$\begin{aligned} \mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)] &\geq \mathbb{E}_{\mathbf{H}_1}[C(\mathbf{H}_1, P)] \\ &\geq \mathbb{E}_{\mathbf{H}} \left[\log \left| \mathbf{I} + \frac{P}{N} \mathbf{H}_1^\dagger \mathbf{H}_1 \right| \right] \\ &= N \mathbb{E}_{\mathbf{H}} \left[\log \left(1 + \frac{P}{N} \lambda_i \right) \right] \\ &\geq N \left(\log \left(\frac{P}{N} \right) + \mathbb{E}_{\mathbf{H}} [\log(\lambda_i)] \right). \end{aligned}$$

where λ_i is an unordered eigenvalue of the Wishart matrix $\mathbf{H}_1 \mathbf{H}_1^\dagger$ and the single-user capacity is lower bounded by transmitting equal power (as opposed to the optimal water-filling power allocation) on each of the N eigenmodes of User 1.

Using Theorem 1 and Jensen's inequality, we can upper bound the sum-rate capacity as:

$$\begin{aligned} \mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)] &\leq M \mathbb{E}_{\mathbf{H}} \left[\log \left(1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2 \right) \right] \\ &\leq M \log \left(1 + \frac{P}{M} \mathbb{E}_{\mathbf{H}} [\|\mathbf{H}\|_{max}^2] \right). \end{aligned}$$

We can also lower bound the sum-rate capacity by choosing $\mathbf{Q}_i = \frac{P}{KN} \mathbf{I}$ in (4) for each user:

$$\begin{aligned} \mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)] &\geq \mathbb{E}_{\mathbf{H}} \left[\log \left| \mathbf{I} + \frac{P}{KN} \mathbf{H}^\dagger \mathbf{H} \right| \right] \\ &= M \mathbb{E}_{\mathbf{H}} \left[\log \left(1 + \frac{P}{KN} \lambda_i \right) \right] \\ &\geq M \left(\log \left(\frac{P}{KN} \right) + \mathbb{E}_{\mathbf{H}} [\log(\lambda_1)] \right) \end{aligned}$$

where λ_i is distributed as an unordered eigenvalue of the $M \times M$ Wishart matrix $\mathbf{H}^\dagger \mathbf{H}$. Using these bounds, as P becomes large, we can upper and lower bound the ratio $\frac{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}$ by $\frac{M}{N}$. It then follows that $\frac{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}$ converges to $\frac{M}{N}$ in the limit of high SNR. This result is intuitively closely related to Theorem 4.

In Fig. 5, the ratio of sum-rate capacity to the TDMA sum-rate (i.e. $\frac{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}$) is plotted for a system with 10 users. The ratio is plotted for $M = 4$ and $N = 1$, $N = 2$, and $N = 4$. In each case the DPC gain converges to $\frac{M}{N}$, though it does so quite slowly for the $N = 1$ case.

B. Large M

In this section we examine the scenario where the number of users (K), number of receive antennas (N), and SNR (P) are fixed but the number of transmit antennas (M) is taken to be large. We will show that the DPC gain converges to K in this case, i.e.:

$$\lim_{M \rightarrow \infty} \frac{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]} = K. \quad (24)$$

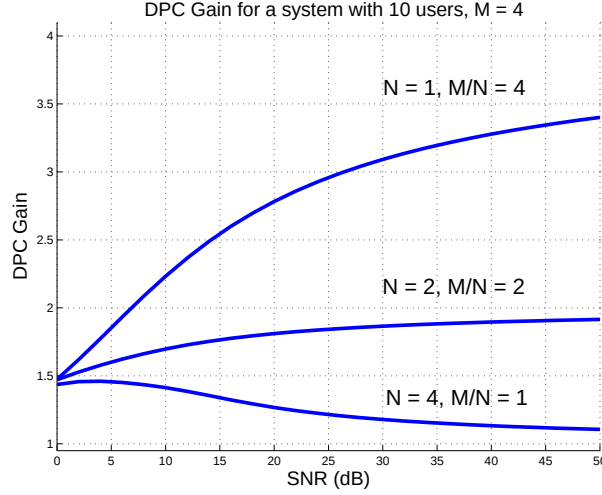


Fig. 5. DPC Gain as a function of SNR for a system with 10 users

As in the previous section, we lower bound the sum rate capacity by choosing $\mathbf{Q}_i = \frac{P}{KN} \mathbf{I}$ in (4) for each user and in each fading state. Since the identity covariance is optimal for point-to-point MIMO channels in Rayleigh fading [12], with the above choice of $\mathbf{Q}_i$ the lower bound is equal to the point-to-point capacity of a KN transmit, M receive MIMO channel where only the receiver has channel knowledge. If the number of receive antennas in this point-to-point link is allowed to become large (i.e. $M \rightarrow \infty$) but the number of transmit antennas in this point-to-point model (KN) is kept fixed, then the capacity of the point-to-point system grows as $KN \log(1 + \frac{MP}{KN})$ [15].

As in the previous section, the TDMA sum-rate is upper bounded as $\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)] \leq N \mathbb{E}_{\mathbf{H}}[\log(1 + \frac{P}{N} \|\mathbf{H}\|_{max}^2)]$. Using standard probability arguments, we can upper bound $\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]$ by $N \log(1 + \frac{P}{N} M(1 + \alpha))$, where α is any strictly positive number.

If we now take the ratio of DPC sum-rate capacity to TDMA sum-rate as $M \rightarrow \infty$, we get

$$\lim_{M \rightarrow \infty} \frac{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}{\mathbb{E}_{\mathbf{H}}[\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]} \geq \lim_{M \rightarrow \infty} \frac{KN \log(1 + \frac{MP}{KN})}{N \log(1 + \frac{P}{N} M(1 + \alpha))} = K. \quad (25)$$

By Theorem 3, this ratio is also upper-bounded by K for all M . Thus, in the limit of many transmit antennas and with a fixed number of receivers, the DPC gain goes to K . Intuitively, as M becomes large, the NK rows of $\mathbf{H}$ become mutually orthogonal because each row is a random vector in $\mathbb{C}^M$. Using TDMA, signaling can be done over N roughly orthogonal dimensions, whereas DPC allows signaling over NK dimensions. Thus, DPC can use K times as many signaling dimensions. Furthermore, the received SNR increases linearly with M . Thus, we are effectively in the high SNR regime when considering asymptotically large M , which implies that the factor of K increase in the number of spatial dimensions gained by using DPC translates to a factor of K increase in rate.

In Figure 6 the DPC gain is plotted as a function of the number of transmit antennas for a system with 3 users, each with 10 dB average SNR. Notice that for both $N = 1$ and $N = 2$, slow convergence to $K = 3$ is observed as

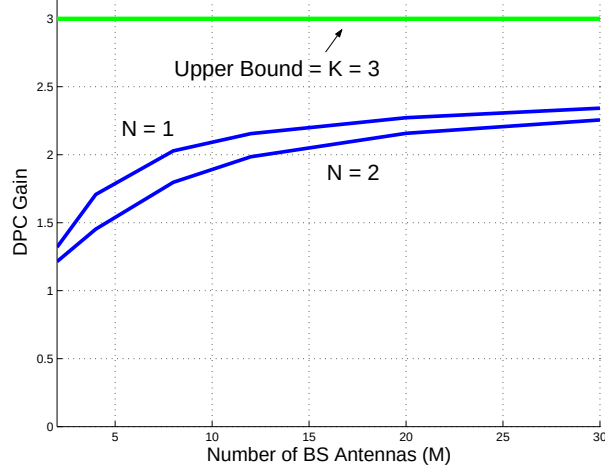


Fig. 6. DPC Gain as a function of M for a system with 3 users at 10 dB

M becomes large.

C. Large K

If the number of antennas and the SNR are kept fixed and the number of users is taken to be large, it is shown in [16] that the dirty paper gain converges to $\frac{M}{N}$, i.e.:

$$\lim_{K \rightarrow \infty} \frac{\mathbb{E}_{\mathbf{H}} [\mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]}{\mathbb{E}_{\mathbf{H}} [\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)]} = \frac{M}{N} \quad (26)$$

More specifically, the authors show that the sum rate capacity and the TDMA sum-rate grow as $M \log \log(K)$ and $N \log \log(K)$, respectively. The intuition in this scenario is that as the number of users grows large, you can find a roughly orthogonal set (of size M or N) of channels to transmit over. Furthermore, the quality of these channels (i.e. the channel gain) grows roughly as $\log(K)$ because the maximum of independent exponential random variables grows logarithmically. Thus, the approximate rate is equal to (# of dimension) $\cdot \log \log(K)$.

D. Numerical Results

In this section we provide plots and analysis of the DPC gain in Rayleigh fading for more realistic system parameters. In Fig. 7, the DPC gain is plotted as a function of the number of users for a system with four transmit antennas and one receive antenna. Plots are provided for different average SNR values. The DPC gain converges to four in the limit of a large number of users (i.e. each SNR curve converges to four), but convergence occurs extremely slow, particularly for the lower SNR values. However, a factor of 2 to 3 increase in sum rate is possible for systems with 20 users and average SNR's ranging from 5 to 20 dB.

In Figure 8 the DPC gain is plotted against SNR for systems with single receive antennas and differing numbers of transmit antennas. Each curve converges to M at very high SNR, and convergence occurs relatively quickly, unlike in the previous figure. Notice that at 0 dB, the DPC gain is larger than two only when there are eight transmit antennas. However, at 10 to 20 dB, gains of 2 to 5 are feasible if four or more antennas are used.

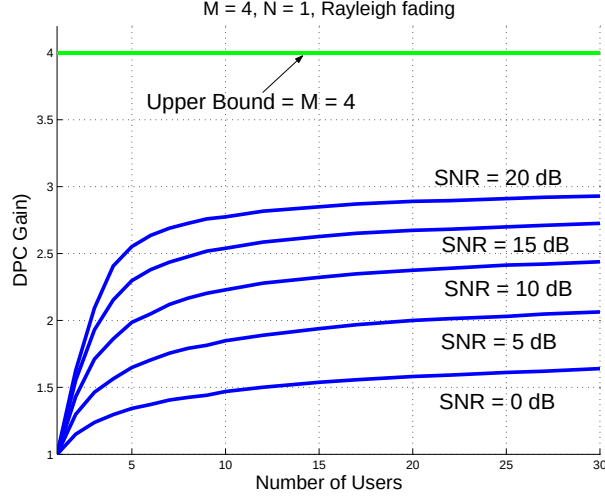


Fig. 7. DPC Gain as a function of # of users for a system with 4 TX antennas and 1 RX antenna

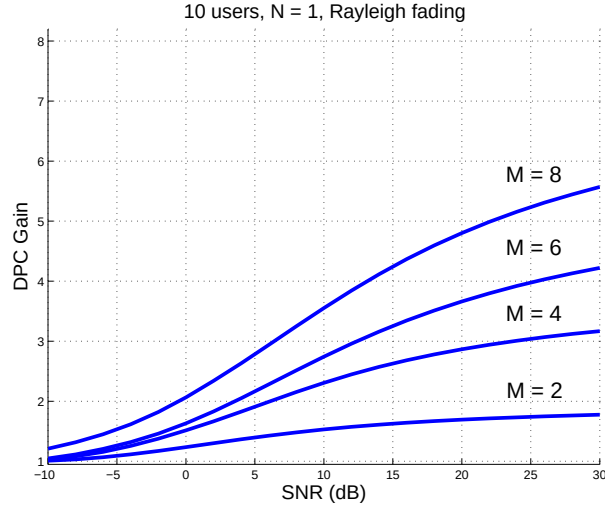


Fig. 8. DPC Gain as a function of SNR for a system with 10 users and 1 RX antenna

In general, note that the largest DPC gain occurs in systems with a large number of users relative to the number of transmit antennas operating at a high average SNR. Cellular systems typically have a large number of users per cell (i.e. 20 - 30), which is large compared to reasonable base station antenna deployments, but SNR's are typically quite low.

VII. RATE REGION BOUNDS

Given that the sum-rate gain of DPC over TDMA can be elegantly upper bounded by $\min(M, K)$, a natural question to ask is whether the entire DPC rate region can be upper bounded by the same factor $\min(M, K)$ times the TDMA rate region, i.e. is $\mathcal{R}_{DPC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \subseteq \min(M, K) \cdot \mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$? We have thus far investigated the gain DPC provides when all users' rates are weighted equally, but such a result would bound

the total gain DPC can provide for *any* weighting of transmission rates.

By single-user capacity bounds, we know that $\mathbf{R} \in \mathcal{R}_{DPC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ implies $\mathbf{R}_i \leq C(\mathbf{H}_i, P)$ for $i = 1, \dots, K$. Thus, $\frac{1}{K} \sum_{i=1}^K \mathbf{R}_i \leq \sum_{i=1}^K \frac{1}{K} C(\mathbf{H}_i, P)$, which by the definition of $\mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ in (5) implies $\frac{1}{K} \mathbf{R} \in \mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$. Thus, it follows that $\mathcal{R}_{DPC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \subseteq K \cdot \mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$. However, it is in fact surprisingly simple to show that it is *not always true* that $\mathcal{R}_{DPC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \subseteq M \cdot \mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$. Consider a single transmit and single receive antenna broadcast channel ($M = N = 1$). As long as each of the channel gains are not equal, the capacity region (which is equal to the DPC region for the scalar broadcast channel) of the scalar broadcast channel is strictly larger than the TDMA region [10], i.e. $\mathcal{R}_{DPC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \supset \mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$. Interestingly, for symmetric channels (i.e. single user capacities of all users are equal $C(\mathbf{H}_1, P) = C(\mathbf{H}_2, P) = \dots = C(\mathbf{H}_K, P)$), the $\min(M, K)$ bound on the DPC gain implies that for any $\mathbf{R}$ in the DPC region, $\sum_{i=1}^K R_i \leq \mathcal{C}_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \leq \min(M, K) \cdot C(\mathbf{H}_1, P)$. Therefore, it follows that $\mathcal{R}_{DPC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \subseteq \min(M, K) \cdot \mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ for symmetric channels. However, this result is not true in general.

VIII. FREQUENCY SELECTIVE BROADCAST CHANNELS

A frequency selective multiple-antenna broadcast channel can be decomposed into a set of parallel, independent flat-fading multiple-antenna broadcast channels [17]. In practical systems, orthogonal frequency division multiplexing (OFDM) can be used to reduce a frequency-selective channel into a finite number of parallel, frequency-flat MIMO broadcast channels (corresponding to each frequency tone). Dirty paper coding could be used separately on each of these frequency tones, or TDMA could be used on each tone (i.e choose only user per tone to transmit to). If DPC is used on each tone, power can optimally be allocated to different tones to meet an average power constraint. If this same per-tone power allocation is used in conjunction with TDMA instead of DPC (transmitting to only the strongest user on each tone), Theorem 3 can be used to upper bound the DPC gain in each tone by $\min(M, K)$. Therefore, the total sum rate achieved using DPC (and optimal allocation of power across tones) is still at most $\min(M, K)$ times larger than the rate achievable using TDMA on each tone. However, if only one user was selected for transmission across all tones, which implies that this user may not be the best user for each tone, the gain of DPC is no longer upper bounded by M , but is still upper bounded by K , the number of users.

IX. TRANSMITTER BEAMFORMING

Transmitter beamforming⁵ is a sub-optimal technique that supports simultaneous transmission to multiple users on a broadcast channel. Each active user is assigned a beamforming direction by the transmitter and multi-user interference is treated as noise. Transmit beamforming is actually quite similar to dirty paper coding, but with DPC some multi-user interference is “pre-subtracted” at the transmitter, thus increasing the rates of some users. In [5] it is shown that transmitter beamforming for the broadcast channel without pre-coding is dual to receiver beamforming in the multiple-access channel without successive interference cancellation. As a result, when $N = 1$, the maximum

⁵Transmitter beamforming is also referred to as SDMA, or space-division multiple access.

sum rate using beamforming can be written as [18]:

$$C_{BF}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) = \max_{\{P_i: \sum_{i=1}^K P_i \leq P\}} \sum_{j=1}^K \log \frac{\left| \mathbf{I} + \sum_{i=1}^K \mathbf{H}_i^\dagger P_i \mathbf{H}_i \right|}{\left| \mathbf{I} + \sum_{i \neq j} \mathbf{H}_i^\dagger P_i \mathbf{H}_i \right|} \quad (27)$$

This optimization cannot be cast in a convex form, and does not lend itself to numerical computation. However, we are able to analytically show that beamforming and DPC are equivalent at low and high SNR.

Theorem 6: If $\mathbf{H}$ has at least M independent rows, beamforming performs as well as DPC in the ratio sense at both asymptotically low and high SNR.

$$\lim_{P \rightarrow \infty} \frac{C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{C_{BF}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)} = \lim_{P \rightarrow 0} \frac{C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{C_{BF}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)} = 1. \quad (28)$$

Proof: By Theorem 1, we have $C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \leq M \log \left(1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2 \right)$. For simplicity, assume that the first M rows of $\mathbf{H}$ are linearly independent. In the proof of Theorem 4, we show that the sum rate achievable using channel inversion is at least as large as $\sum_{i=1}^M \log(1 + \alpha_i^2 \frac{P}{M})$. Channel inversion is a particular method of transmitter beamforming (zero-forcing beamforming), and thus $C_{BF}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \geq \sum_{i=1}^M \log(1 + \alpha_i^2 \frac{P}{M})$. Therefore we have:

$$\begin{aligned} \lim_{P \rightarrow \infty} \frac{C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{C_{BF}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)} &\leq \lim_{P \rightarrow \infty} \frac{M \log \left(1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2 \right)}{\sum_{i=1}^M \log(1 + \alpha_i^2 \frac{P}{M})} \\ &= 1 \end{aligned}$$

The low SNR result follows directly from Theorem 5 and the fact that $C_{BF}(\mathbf{H}, P) \geq C_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$. ■

The equivalence of transmitter beamforming and DPC at high SNR follows from the fact that both DPC and transmitter beamforming can use $\min(M, NK)$ signaling dimensions. However, the use of dirty paper coding reduces interference seen at the receivers and therefore improves the quality of each of the signaling dimensions, leading to an increase in sum rate at finite SNR. Thus, an interesting open problem is to analytically bound the gain that DPC provides over transmitter beamforming. We conjecture that the ratio $\frac{C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{C_{BF}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}$ is bounded by a constant ($< M$) independent of the number of antennas and the channel $\mathbf{H}$ for all P , but we have been unable to prove this due to the difficulty of working with the beamforming sum-rate expression in (27). Viswanathan and Venkatesan [19] recently characterized the performance of downlink beamforming and dirty paper coding as M and K both grow to infinity at some fixed ratio $\frac{M}{K} = \alpha$. In this asymptotic regime, the ratio $\frac{C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{C_{BF}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}$ is bounded by 2 for all values of α and P .

X. BOUND ON SUM-RATE GAIN OF SUCCESSIVE DECODING FOR UPLINK

Successive decoding is a capacity-achieving scheme for the multiple-access channel (uplink) in which multiple users simultaneously transmit to the base station and the receiver successively decodes and subtracts out the signals of different users. This technique achieves the sum-rate capacity of the MIMO MAC [10, Chapter 14], but is difficult to implement in practice. The sum-rate capacity of the MAC with transmitter power constraints $\mathbf{P} = (P_1, \dots, P_K)$

is given by:

$$\mathcal{C}_{MAC}(\mathbf{H}_1, \dots, \mathbf{H}_K, \mathbf{P}) = \max_{\{\text{Tr}(\mathbf{Q}_i) \leq P_i \forall i\}} \log \left| \mathbf{I} + \sum_{i=1}^K \mathbf{H}_i^\dagger \mathbf{Q}_i \mathbf{H}_i \right|.$$

Notice that the sum-rate capacity of the MAC is identical to the BC sum-rate capacity expression in (4) except that the MAC expression has *individual* power constraints instead of a sum constraint.

Using the proof technique of Theorem 1 on the dual uplink (K transmitters with N antennas each and a single receiver with M antennas) modified for individual power constraints $\mathbf{P} = (P_1, \dots, P_K)$, it is easy to see that the following holds:

$$\mathcal{C}_{MAC}(\mathbf{H}_1, \dots, \mathbf{H}_K, \mathbf{P}) \leq M \log \left(1 + \frac{\sum_{i=1}^K P_i \|\mathbf{H}_i\|^2}{M} \right). \quad (29)$$

A sub-optimal transmission scheme is to allow only one user to transmit at a time. Since each user in the uplink has an individual power constraint, users are allocated orthogonal time slots in which they transmit. Thus, the TDMA rate region is defined as:

$$\mathcal{R}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, \mathbf{P}) \triangleq \bigcup_{\alpha_i \geq 0, \sum_{i=1}^K \alpha_i = 1} \left(\alpha_1 C \left(\mathbf{H}_1, \frac{P_1}{\alpha_1} \right), \dots, \alpha_K C \left(\mathbf{H}_K, \frac{P_K}{\alpha_K} \right) \right).$$

As before, the TDMA sum-rate is defined as the maximum sum of rates in this region. As used in the proof of Theorem 2, for each user $C(\mathbf{H}_i, \frac{P_i}{\alpha_i}) \geq \log(1 + \frac{P_i}{\alpha_i} \|\mathbf{H}_i\|^2)$ for any α_i . Thus,

$$\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, \mathbf{P}) \geq \max_{\alpha_i \geq 0, \sum_{i=1}^K \alpha_i = 1} \sum_{i=1}^K \alpha_i \log \left(1 + \frac{P_i}{\alpha_i} \|\mathbf{H}_i\|^2 \right).$$

The RHS of this expression corresponds to the TDMA region of a *scalar* MAC with channel gains $\|\mathbf{H}_1\|, \dots, \|\mathbf{H}_K\|$. It is easy to verify that this expression is maximized by choosing $\alpha_i = \frac{P_i \|\mathbf{H}_i\|^2}{\sum_{j=1}^K P_j \|\mathbf{H}_j\|^2}$. With this choice of α_i , we get the following upper bound:

$$\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, \mathbf{P}) \geq \log \left(1 + \sum_{i=1}^K P_i \|\mathbf{H}_i\|^2 \right). \quad (30)$$

Combining (29) and (30) we get $\frac{\mathcal{C}_{MAC}(\mathbf{H}_1, \dots, \mathbf{H}_K, \mathbf{P})}{\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, \mathbf{P})} \leq M$. As before, the single-user capacity of each user also upper bounds this ratio by K . Thus, we finally get

$$\boxed{\frac{\mathcal{C}_{MAC}(\mathbf{H}_1, \dots, \mathbf{H}_K, \mathbf{P})}{\mathcal{C}_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, \mathbf{P})} \leq \min(M, K)} \quad (31)$$

or that performing optimal successive decoding at the base station offers a gain of at most $\min(M, K)$ over TDMA.

XI. CONCLUSION

We have established that the sum-rate capacity of a multiple-antenna broadcast channel, achievable using dirty paper coding, is at most $\min(M, K)$ times larger than the maximum achievable sum rate using TDMA, where M

is the number of transmit antennas (base station antennas) and K is the number of users. This bound applies at any SNR and for any number of receive antennas. For Rayleigh fading channels, the bound tightens to $\min(\frac{M}{N}, K)$ at high SNR, for a large number of transmit antennas, or for a large number of users. Using the same techniques for the uplink, we found that the performance gain using successive decoding on the uplink versus TDMA is also upper bounded by $\min(M, K)$, where M is the number of receive antennas (at the base station) and K is the number of mobiles (i.e. transmitters). Thus, it seems that for systems with many users that operate at relatively high SNR's, dirty paper coding can provide a significant performance gain if additional antennas are added only at the base station without the extra expense associated with adding additional antennas at each mobile. In contrast, if the number of mobile antennas is the same as the number of base station antennas or if the system is operating at low SNR, the benefit of using DPC on the downlink or successive decoding on the uplink is rather limited.

XII. ACKNOWLEDGMENT

The authors wish to thank Jerry Foschini for his assistance with this work.

APPENDIX

I. PROOF OF THEOREM 4

The basic premise is to show that $C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ grows as $\min(M, NK) \log(P)$ and $C_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ grows as $N \log(P)$. We first consider a channel with $M < NK$. Note that by Theorem 1, $C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \leq M \log(1 + \frac{P}{M} \|\mathbf{H}\|_{max}^2)$. Furthermore, we form a matrix (denoted by $\mathbf{H}^{M,M}$) consisting of any set of M linearly independent rows of $\mathbf{H}$. By the linear independence of the rows, $\mathbf{H}^{M,M}$ is strictly positive definite. Therefore, we can invert the channel at the transmitter to give M independent and parallel non-zero channels, though there is a power penalty involved in channel inversion. The result is M independent and parallel channels with strictly non-zero channel gains $\alpha_1, \dots, \alpha_M$. Thus, we have $C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \geq \sum_{i=1}^M \log(1 + \alpha_i^2 \frac{P}{M})$, because allocating equal power on the parallel channels is sub-optimal.

Without loss of generality, assume $\mathbf{H}_1$ has full row rank. Then a lower bound to $C_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ is arrived at by $C_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \geq C(\mathbf{H}_1, P) \geq \sum_{i=1}^N \log(1 + \lambda_i \frac{P}{N})$, where $(\lambda_1, \dots, \lambda_N)$ are the eigenvalues of $\mathbf{H}_1 \mathbf{H}_1^\dagger$. Similar to Theorem 1, we can upper bound $C_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$ by assuming there exists a user with N eigenmodes with eigenvalues equal to $\|\mathbf{H}\|_{max}$: $C_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) \leq N \log(1 + \|\mathbf{H}\|_{max}^2 \frac{P}{N})$. Combining these bounds, we can establish that $\lim_{P \rightarrow \infty} \frac{C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}{C_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)}$ is bounded above and below by $\frac{M}{N}$.

If $KN \leq M$ (i.e. $K \leq \frac{M}{N}$), we can similarly show that DPC grows as $KN \log(P)$ while TDMA grows as $N \log(P)$. Thus, the DPC gain is asymptotically equal to K .

II. PROOF OF THEOREM 5

Without loss of generality, assume that $|\mathbf{H}_1| \geq |\mathbf{H}_2| \cdots \geq |\mathbf{H}_K|$. First notice that on the dual MAC we have:

$$\log \left| \mathbf{I} + \sum_{i=1}^K \mathbf{H}_i^\dagger \mathbf{Q}_i \mathbf{H}_i \right| = I(X_1, \dots, X_K; Y) \quad (32)$$

$$= \sum_{i=1}^K I(X_i; Y | X_1, \dots, X_{i-1}) \quad (33)$$

$$\leq \sum_{i=1}^K I(X_i; Y | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_K) \quad (34)$$

$$= \sum_{i=1}^K \log \left| \mathbf{I} + \mathbf{H}_i^\dagger \mathbf{Q}_i \mathbf{H}_i \right| \quad (35)$$

where we have used the chain rule for mutual information in (33) and the fact that $X_1, \dots, X_K$ are independent to get $I(X_i; Y | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_K) \geq I(X_i; Y | X_1, \dots, X_{i-1})$ in (34). Thus we can upper bound the sum capacity of the dual MIMO MAC by the sum capacity of a multiple-access channel where each transmitter has an independent channel ($\mathbf{H}_i^\dagger$) to the receiver. By diagonalizing each $\mathbf{H}_i$, we can write the sum capacity of this upper bound as:

$$\max_{P_{i,j} \geq 0: \sum_{i,j} P_{i,j} \leq P} \sum_{i,j} \log(1 + P_{i,j} \lambda_{i,j}) \quad (36)$$

where $\lambda_{i,j}$ is the j -th eigenvalue of $\mathbf{H}_i \mathbf{H}_i^\dagger$. Assume that $\lambda_{i,j} \geq \lambda_{i,k}$ for all $k > j$. Notice that the optimal power allocation maximizing (36) is found by the standard water-filling procedure for parallel Gaussian channels [10, Chapter 10.4].

We must separately consider two different cases: $|\mathbf{H}_1| > |\mathbf{H}_2|$ and $|\mathbf{H}_1| = |\mathbf{H}_2|$. First consider $|\mathbf{H}_1| > |\mathbf{H}_2|$, which implies $\lambda_{1,1} > \lambda_{2,1}$. From the water-filling procedure, we know that for $P \leq \frac{1}{\lambda_{1,1}} - \frac{1}{\lambda_{2,1}}$, there is not enough power to fill any of the channels of Users $2, \dots, K$. Therefore, the maximum in (36) is achieved by only allocating power to User 1, i.e. TDMA is optimal, for small enough P . Thus, for $P \leq \frac{1}{\lambda_{1,1}} - \frac{1}{\lambda_{2,1}}$, we have $C_{BC}(\mathbf{H}_1, \dots, \mathbf{H}_K, P) = C_{TDMA}(\mathbf{H}_1, \dots, \mathbf{H}_K, P)$.

Now consider the scenario where $|\mathbf{H}_1| = |\mathbf{H}_2| = \dots = |\mathbf{H}_L|$, i.e. where L users have the same largest eigenvalue. Let m_i be the multiplicity of the largest eigenvalue of the i -th user, and assume $\sum_{i=1}^L m_i = J$. Then for $P \leq J(\frac{1}{\lambda_{1,1}} - \frac{1}{\lambda^*})$, where λ^* is the second largest eigenvalue amongst all users, the maximum in (36) is achieved by allocating equal power to the J eigenmodes with the largest eigenvalue. The corresponding capacity is $J \log(1 + |\mathbf{H}_1|^2 \frac{P}{J})$. In this case, note that

$$\lim_{P \rightarrow 0} \frac{J \log(1 + |\mathbf{H}_1|^2 \frac{P}{J})}{\log(1 + |\mathbf{H}_1|^2 P)} = 1. \quad (37)$$

Thus, in the limit of small P , the ratio of the sum capacity of the upper bound to the TDMA capacity goes to one.

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